

SAINT-MICHEL-DES-SAINTS, QC, Canada

WMO: 717370

Lat: 46.816N

Lon: 74.094W

Elev: 430

StdP: 96.27

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.1	(c) -27.2	(d) -33.4	(e) 0.2	(f) -29.4	(g) -30.5	(h) 0.2	(i) -26.3	(j) 7.3	(k) -11.5	(l) 6.4	(m) -12.2	(n) 0.6	(o) 280	(p) 0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	27.9	18.9	26.3	18.4	24.7	17.5	21.0	25.2	20.0	23.6	19.2	22.9	3.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	15.2	22.8	18.8	14.4	22.0	17.8	13.4	21.1	63.1	25.4	59.5	23.4	56.5	22.9	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.2	(b) 5.4	(c) 4.8	(e) -34.2	(f) 30.7	(g) 2.0	(h) 1.3	(i) -35.6	(j) 31.7	(k) -36.7	(l) 32.5	(m) -37.8	(n) 33.2	(o) -39.3	(p) 34.2		
(a) 6.2	(b) 5.4	(c) 4.8	(e) -34.8	(f) 22.7	(g) 3.1	(h) 0.7	(i) -37.0	(j) 23.2	(k) -38.8	(l) 23.6	(m) -40.6	(n) 24.0	(o) -42.8	(p) 24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-12.9	-11.8	-5.7	1.5	10.3	14.4	17.7	16.5	12.4	5.5	-1.7	-9.2
	DBStd	12.07	7.50	6.85	6.84	5.19	4.82	3.63	2.95	2.93	4.52	4.56	5.68	7.08
	HDD10.0	3251	710	610	488	257	55	4	0	0	26	153	353	596
	HDD18.3	5599	968	843	746	504	251	128	48	71	184	399	602	854
	CDD10.0	754	0	0	1	3	66	135	238	202	98	12	0	0
	CDD18.3	62	0	0	0	0	4	9	28	15	6	0	0	0
	CDH23.3	661	0	0	0	1	73	112	302	122	51	1	0	0
	CDH26.7	105	0	0	0	0	14	16	55	13	7	0	0	0
	WSAvg	2.0	2.1	2.2	2.3	2.4	2.0	1.9	1.8	1.6	1.6	1.9	2.0	1.9
	PrecAvg	1028	78	58	64	72	83	102	107	103	103	93	82	82
(15) Precipitation	PrecMax	1234	151	148	122	151	132	170	175	191	229	146	150	130
	PrecMin	872	32	26	11	15	46	44	63	47	42	25	26	35
	PrecStd	84	27	25	27	29	28	33	30	32	43	34	27	27
	PrecStd	84	27	25	27	29	28	33	30	32	43	34	27	27
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	6.1	16.1	19.6	28.4	28.5	30.6	28.1	27.8	21.4	13.9	6.2
		MCWB	7.2	3.1	11.8	13.1	17.2	18.4	21.0	19.4	20.6	14.7	10.4	5.4
	2%	DB	3.3	2.1	9.0	15.6	25.3	26.5	28.2	26.3	24.6	17.8	10.6	3.7
		MCWB	2.1	0.7	4.3	8.8	15.2	17.3	19.5	19.2	18.8	12.9	7.0	3.1
	5%	DB	0.3	0.2	6.3	12.7	22.7	24.4	26.5	24.6	22.5	15.5	8.1	1.4
		MCWB	-0.3	-1.0	2.2	7.2	13.9	16.5	18.8	18.1	17.6	12.0	5.7	0.7
	10%	DB	-2.0	-1.1	3.9	9.9	19.9	22.3	24.9	22.9	20.6	13.0	5.8	0.2
		MCWB	-2.7	-2.1	0.8	5.1	12.5	15.1	18.1	17.7	16.9	10.6	4.2	-0.5
	0.4%	WB	7.3	5.2	12.3	14.1	19.4	21.2	22.6	21.5	21.3	16.2	11.6	5.6
		MCDB	7.6	5.6	16.2	18.9	24.8	26.1	27.5	25.4	25.4	19.1	13.0	6.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	2.4	1.0	5.6	10.9	17.8	19.5	21.4	20.2	19.9	14.3	8.0	3.1
		MCDB	2.9	1.8	7.5	13.3	22.6	23.5	25.7	23.8	22.7	16.9	9.9	3.7
	5%	WB	-0.3	-0.5	3.0	7.8	16.1	18.2	20.3	19.3	18.8	12.6	5.9	0.8
		MCDB	0.4	0.3	5.8	11.2	19.6	21.6	23.9	22.9	21.6	15.0	7.4	1.2
	10%	WB	-2.7	-1.8	1.2	5.7	14.2	16.8	19.3	18.4	17.5	10.7	4.3	-0.4
		MCDB	-2.0	-1.0	3.4	9.5	17.5	20.7	23.3	21.8	20.2	12.4	5.5	0.1
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	11.5	11.8	10.9	12.6	11.9	12.1	11.2	11.4	9.2	8.3	8.7
		MCDBR	10.4	10.1	14.6	16.1	17.5	15.1	14.5	13.7	13.3	14.2	11.5	8.0
		MCWBR	10.0	9.1	9.9	9.9	8.5	7.2	7.2	7.3	7.9	9.6	8.8	7.8
	5% WB	MCDBR	10.5	10.2	12.3	13.5	13.2	11.0	12.1	11.2	11.7	12.5	9.8	8.1
		MCWBR	10.3	9.5	8.9	9.7	7.6	6.6	7.0	6.4	7.8	9.4	8.3	8.0
(40) Clear-Sky Solar Irradiance	taub		0.252	0.269	0.298	0.345	0.356	0.381	0.381	0.372	0.352	0.321	0.291	0.257
	taud		2.315	2.256	2.253	2.336	2.419	2.350	2.361	2.390	2.439	2.530	2.501	2.361
	Ebn at Noon		852	906	927	909	909	883	878	873	861	843	804	802
	Edh at Noon		73	97	114	117	112	121	118	110	96	75	63	63
	RadAvg		1.34	2.29	3.66	4.60	5.10	5.55	5.41	4.75	3.69	2.13	1.24	0.94
(45) All-Sky Solar Radiation	RadStd		0.21	0.32	0.35	0.39	0.51	0.41	0.53	0.24	0.33	0.21	0.17	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page